Notes: Barberis, Shleifer, and Vishny (JFE, 1998)

A Model of Investor Sentiment

Contents

1 Big picture

- **Question.** Can one parsimonious behavioral model explain two facts that seem to point in opposite directions: short-run underreaction to news and long-run overreaction to strings of similar news?
- **Paper type.** This is a theory paper, not a new-data empirical paper. The empirical facts are taken from the anomaly literature on post-earnings-announcement drift, momentum, reversals, and value versus glamour returns. The contribution is a model of investor beliefs that can generate these facts together.
- **Core idea.** True earnings follow a random walk, but the representative investor does not know this. Instead, he thinks earnings are generated by one of two regimes: a *mean-reverting* regime and a *trending* regime. He updates Bayesianly over the wrong model.
- **Psychological foundation.** The model is motivated by two judgment biases. *Conservatism* makes investors update too little after isolated news. *Representativeness* makes investors over-extrapolate after a long string of similar news.
- **How underreaction appears.** After one positive earnings shock, the investor often treats the shock as temporary because he puts high probability on the mean-reverting regime. Price therefore rises too little. If earnings are actually a random walk, the next period's expected return is positive.
- **How overreaction appears.** After many positive shocks in a row, the investor shifts probability toward the trending regime. He then expects the positive pattern to continue. Because the true process is a random walk, the stock becomes overpriced and subsequent average returns are low.
- **Why the model is interesting.** The same belief system produces both continuation and reversal. The investor is not irrational in every step: conditional on his perceived model, he uses Bayes' rule. The error is that the perceived model omits the true random-walk process.
- **Main result.** For a nontrivial range of parameter values, the model generates both short-horizon underreaction and long-horizon overreaction. The paper verifies this analytically through sufficient conditions and numerically through simulations.
- **Simulation message.** Artificial data from the model reproduce the sign pattern of classic anomalies: positive return continuation after short earnings or return histories, negative return reversal after longer histories, and higher future returns for high earnings-price stocks.
- **Takeaway.** Investor sentiment can be modeled as biased belief formation. Prices deviate from fundamental value not because investors ignore information, but because they use the wrong mental model to interpret information.

2 Evidence and psychological inputs (key objects)

2.1 The empirical facts the model is built to explain

The paper starts from two broad empirical regularities.

Underreaction. Prices incorporate some information too slowly. A simple way to express this is

$$\mathbb{E}(r_{t+1} \mid z_t = G) > \mathbb{E}(r_{t+1} \mid z_t = B), \tag{1}$$

where $z_t = G$ means good news at time t and $z_t = B$ means bad news. If good news predicts high future returns, then price did not rise enough when the news arrived.

Important examples include:

- post-earnings-announcement drift: stocks with high standardized unexpected earnings continue to outperform stocks with low standardized unexpected earnings;
- momentum: stocks with high recent returns tend to keep outperforming over roughly six- to twelve-month horizons;
- event-study drift after corporate announcements such as repurchases, dividend changes, stock splits, and equity issues.

Overreaction. Prices can move too far after a long series of similar signals. A compact definition is

$$\mathbb{E}(r_{t+1} \mid z_t = G, z_{t-1} = G, \ldots, z_{t-j} = G) < \mathbb{E}(r_{t+1} \mid z_t = B, z_{t-1} = B, \ldots, z_{t-j} = B), \tag{2}$$

for a sufficiently long string length j. After a long sequence of good news, investors become too optimistic and future returns are low. After a long sequence of bad news, investors become too pessimistic and future returns are high.

Important examples include:

- long-run reversals after extreme past returns;
- value stocks outperforming glamour stocks;
- low market-to-book or high earnings-price firms earning higher subsequent returns than high-valuation growth firms.

Economic message. The facts are hard to fit into a single model because underreaction and overreaction seem opposite. Barberis, Shleifer, and Vishny solve this by making the investor's reaction depend on the *pattern* of the news.

2.2 Conservatism

Conservatism is the tendency to update beliefs in the right direction but by too little. In this paper, it maps naturally into underreaction.

Interpretation in asset pricing. A single positive earnings surprise may be statistically informative about the level of future earnings. But a conservative investor treats it as partly transitory. Price rises, but not enough. Later returns are positive as the information is gradually incorporated.

How the model encodes it. Conservatism is represented by the investor's belief in a mean-reverting earnings regime. In that regime, good news is expected to be followed by bad news, and bad news is expected to be followed by good news.

2.3 Representativeness

Representativeness is the tendency to see a pattern as typical of a broader type even when the sample is small. In finance, investors may see a firm with several years of strong growth and infer that it is a genuine growth firm whose future earnings will keep trending upward.

Interpretation in asset pricing. A long string of good earnings makes investors too optimistic. They extrapolate the trend and overprice the stock. When the trend fails to continue, future returns are low.

How the model encodes it. Representativeness is represented by the investor's belief in a trending earnings regime. In that regime, good news is expected to be followed by more good news, and bad news by more bad news.

2.4 Strength versus weight

The paper draws on Griffin and Tversky's distinction between the *strength* and *weight* of evidence.

- **Strength** means salience, extremity, and how persuasive the evidence looks on the surface.
- **Weight** means statistical reliability, such as sample size and true predictive content.

A single earnings announcement may have high statistical weight but low strength, so investors underreact. A visually striking pattern of several similar earnings announcements may have high strength but low statistical weight, so investors overreact.

Economic message. The model is a formal way to translate strength-versus-weight psychology into a pricing model.

2.5 What is being measured conceptually?

The model focuses on the investor's posterior probability that the world is currently in the mean-reverting regime. This probability is denoted q_t:

$$q_t = \mathbb{P}(s_t = 1 \mid y_t, y_{t-1}, q_{t-1}), \tag{3}$$

where $s_t = 1$ is the mean-reverting regime and $s_t = 2$ is the trending regime.

The key state variable is therefore not just whether the latest earnings shock is good or bad. It is the investor's belief about the *type of earnings process* generating the shock.

3 Models and theoretical specifications (all equations + interpretation)

3.1 True earnings process

True earnings follow a random walk:

$$N_t = N_{t-1} + y_t, \tag{4}$$

where the earnings shock is binary:

$$y_t \in \{+y, -y\}, \qquad \mathbb{P}(y_t = +y) = \mathbb{P}(y_t = -y) = \frac{1}{2}. \tag{5}$$

All earnings are paid out as dividends.

Interpretation. If investors knew the true random-walk process, the level of current earnings would be the best forecast of future earnings. With risk neutrality and constant discount rate, returns would not be predictable from past earnings patterns.

3.2 Investor's perceived Model 1: mean reversion

The investor believes that one possible regime is mean reversion. Its transition matrix is

$$M_1 = \begin{array}{c|cc} & y_{t+1} = +y & y_{t+1} = -y \\ \hline y_t = +y & \pi_L & 1 - \pi_L \\ y_t = -y & 1 - \pi_L & \pi_L \end{array}, \qquad 0 < \pi_L < \frac{1}{2}. \tag{6}$$

Interpretation. Under Model 1, a positive shock is more likely to be reversed than repeated because $1 - \pi_L > \pi_L$. This is the formal version of conservatism: investors think individual shocks are too temporary.

3.3 Investor's perceived Model 2: trending earnings

The second possible regime is a trend regime:

$$M_2 = \begin{array}{c|cc} & y_{t+1} = +y & y_{t+1} = -y \\ \hline y_t = +y & \pi_H & 1 - \pi_H \\ y_t = -y & 1 - \pi_H & \pi_H \end{array}, \qquad \frac{1}{2} < \pi_H < 1. \tag{7}$$

Interpretation. Under Model 2, a positive shock is more likely to be followed by another positive shock. This is the formal version of representativeness: after a string of good news, investors think they have identified a persistent growth type.

3.4 Regime-switching process

The investor believes the world switches between the two regimes according to

$$\Lambda = \begin{array}{c|cc} & s_{t+1} = 1 & s_{t+1} = 2 \\ \hline s_t = 1 & 1 - \lambda_1 & \lambda_1 \\ s_t = 2 & \lambda_2 & 1 - \lambda_2 \end{array}, \tag{8}$$

with

$$\lambda_1 + \lambda_2 < 1. \tag{9}$$

The unconditional probability of regime 1 in the investor's perceived model is

$$\mathbb{P}(s_t = 1) = \frac{\lambda_2}{\lambda_1 + \lambda_2}. \tag{10}$$

Interpretation. Small values of λ_1 and λ_2 mean that regimes are persistent. If $\lambda_1 < \lambda_2$, the investor expects the mean-reverting regime to occur more often than the trending regime.

3.5 Bayesian updating over the wrong model

Let

$$a_t = (1 - \lambda_1)q_t + \lambda_2(1 - q_t), \tag{11}$$
$$b_t = \lambda_1 q_t + (1 - \lambda_2)(1 - q_t). \tag{12}$$

Here a_t is the investor's prior probability that $s_{t+1} = 1$ before seeing y_{t+1}, and b_t is the prior probability that $s_{t+1} = 2$.

If the new earnings shock has the same sign as the previous shock, Bayes' rule gives

$$q_{t+1}^{\text{same}} = \frac{a_t \pi_L}{a_t \pi_L + b_t \pi_H}. \tag{13}$$

Because $\pi_H > \pi_L$, observing two shocks with the same sign lowers q_{t+1}: the investor shifts toward the trending regime.

If the new earnings shock reverses sign, Bayes' rule gives

$$q_{t+1}^{\text{reverse}} = \frac{a_t(1 - \pi_L)}{a_t(1 - \pi_L) + b_t(1 - \pi_H)}. \tag{14}$$

Because $1 - \pi_L > 1 - \pi_H$, a reversal raises q_{t+1}: the investor shifts toward the mean-reverting regime.

Economic interpretation. The investor is Bayesian inside his own model. The behavioral element is not arithmetic error. It is model error: the investor never gives positive probability to the true random walk.

3.6 Price function

The representative investor is risk neutral and discounts at rate δ. The price is the present value of expected future earnings:

$$P_t = \mathbb{E}_t \left[\frac{N_{t+1}}{1 + \delta} + \frac{N_{t+2}}{(1 + \delta)^2} + \cdots \right]. \tag{15}$$

If the investor knew the true random-walk process, then $\mathbb{E}_t(N_{t+j}) = N_t$ and the price would be

$$P_t^* = \frac{N_t}{\delta}. \tag{16}$$

The paper's Proposition 1 shows that, under the investor's perceived regime-switching model,

$$P_t = \frac{N_t}{\delta} + y_t(p_1 - p_2 q_t), \tag{17}$$

where p_1 and p_2 depend on $\pi_L, \pi_H, \lambda_1, \lambda_2$.

Interpretation. Mispricing is summarized by the second term:

$$P_t - P_t^* = y_t(p_1 - p_2 q_t). \tag{18}$$

For a positive shock, price is above fundamental value if $p_1 - p_2 q_t > 0$ and below fundamental value if $p_1 - p_2 q_t < 0$. Thus the same positive shock can create underreaction or overreaction depending on q_t.

3.7 Underreaction in the model

The paper studies price changes, using the condition

$$\mathbb{E}_t(P_{t+1} - P_t \mid y_t = +y) - \mathbb{E}_t(P_{t+1} - P_t \mid y_t = -y) > 0. \tag{19}$$

Interpretation. A positive shock predicts a higher next-period price change than a negative shock. This is the model counterpart of post-news drift and momentum.

Underreaction occurs when the average q_t after ordinary news is high enough that the investor puts substantial weight on mean reversion. A positive earnings shock is then treated as partly temporary, and price does not rise enough.

3.8 Overreaction in the model

For a sufficiently long streak, overreaction means

$$\mathbb{E}_t(P_{t+1} - P_t \mid y_t = y_{t-1} = \cdots = y_{t-j} = +y) \tag{20}$$
$$- \mathbb{E}_t(P_{t+1} - P_t \mid y_t = y_{t-1} = \cdots = y_{t-j} = -y) < 0. \tag{21}$$

Interpretation. After a long positive streak, the next average price change is lower than after a long negative streak. This captures long-horizon reversal.

A long positive streak lowers q_t because same-sign shocks are interpreted as evidence for the trending regime. Then $p_1 - p_2 q_t$ can become positive, making the stock overpriced relative to random-walk fundamentals.

3.9 Sufficient conditions

The paper's Proposition 2 gives sufficient conditions for the same price function to exhibit both underreaction and overreaction:

$$\underline{k} p_2 < p_1 < \bar{k} p_2, \qquad p_2 \geq 0, \tag{22}$$

where $\underline{k}$ and $\bar{k}$ are positive constants determined by $\pi_L, \pi_H, \lambda_1, \lambda_2$.

Economic interpretation. The ratio p_1/p_2 must lie between a low-q threshold and an average/high-q threshold. If it is too low, even long streaks do not generate overreaction. If it is too high, even isolated news does not generate underreaction. The interesting region is the middle region, where the investor underreacts to isolated news but overreacts to patterns.

3.10 Simulation design

The paper simulates artificial data using parameters that lie in the region where both effects occur:

$$\lambda_1 = 0.1, \qquad \lambda_2 = 0.3, \qquad \pi_L = \frac{1}{3}, \qquad \pi_H = \frac{3}{4}. \tag{23}$$

The simulation uses 2,000 independent firms and six annual earnings realizations per firm. True earnings are generated by the random walk, while prices are generated by the investor's perceived regime-switching model.

Purpose. The simulations ask whether the model reproduces the qualitative signs of familiar return patterns: short-run continuation, long-run reversal, and value-stock outperformance.

4 Mechanism, discipline, and limitations

4.1 Why one shock leads to underreaction

Suppose the latest earnings shock is positive. If the investor puts high probability on Model 1, he expects reversal. The current price is therefore below the random-walk fundamental value. Because the true earnings process is a random walk, the next shock is not biased toward reversal. On average, the stock earns a positive subsequent return.

Economic message. Underreaction is not caused by investors ignoring the signal. It is caused by investors interpreting an informative signal as too temporary.

4.2 Why a long streak leads to overreaction

After a long series of positive earnings shocks, the investor puts more weight on Model 2. He now expects future earnings shocks to continue in the same direction. Since true earnings are a random walk, the expected continuation is too optimistic. The price rises above fundamental value, creating low subsequent returns.

Economic message. Overreaction is not a separate force from underreaction. It comes from the same updating rule applied to a different information pattern.

4.3 What disciplines the model

Although the paper is not an empirical identification paper, the model is disciplined in several ways.

- **Psychological discipline.** The two regimes map to documented biases: conservatism and representativeness.

- **Joint-anomaly discipline.** The model must produce both short-run continuation and long-run reversal, not just one anomaly.
- **Parameter discipline.** The paper does not choose a single knife-edge parameterization. It shows nontrivial parameter regions where both effects occur.
- **Simulation discipline.** The model is fed random-walk earnings and asked to reproduce the sign patterns of earnings sorts, return sorts, and earnings-price sorts.

4.4 Role of limited arbitrage

The model itself has one representative investor, so sentiment directly affects prices. The paper does not build a full arbitrage sector into the model. Instead, it relies on the broader limited-arbitrage literature: mispricing can persist because betting against sentiment is risky, especially when sentiment can become more extreme before correcting.

Interpretation. The model explains the *form* of sentiment-driven mispricing. It does not separately prove why rational arbitrageurs fail to eliminate it.

4.5 Strengths

- It gives a unified explanation for both momentum-like and reversal-like patterns.
- It uses a small number of states and parameters.
- It formalizes psychological evidence without making investors randomly irrational.
- It generates value effects even though the representative investor is risk neutral, so the premium is not a compensation-for-risk result inside the model.

4.6 Limitations

- **Wrong model is fixed.** The investor never learns the true random walk, even after seeing a long history.
- **No microfoundation for model choice.** The paper motivates the two perceived regimes psychologically, but does not derive them from deeper cognitive primitives.
- **Representative-agent simplification.** Heterogeneous beliefs and trading among investors are not modeled.
- **Limited arbitrage is external.** The model assumes sentiment can affect price; it does not endogenize arbitrageurs' constraints.
- **Not all event-study facts fit perfectly.** The paper notes that the model has difficulty explaining underreaction to quarterly earnings surprises for extreme growth stocks unless extended.
- **Strength and weight are not measured independently.** The model needs a sharper way to classify events before seeing returns.

4.7 Relation to alternative behavioral models

Daniel, Hirshleifer, and Subrahmanyam develop a related model based on overconfidence and self-attribution. Barberis, Shleifer, and Vishny instead rely on conservatism and representativeness. Both approaches try to explain why prices may first underreact and then overreact.

Economic message. The paper is not saying this is the only behavioral mechanism. It shows that one disciplined belief-formation mechanism is sufficient to generate a broad set of anomalies.

5 Tables and figures

5.1 Table 1: belief updating in a sample earnings path

Read. Table 1 shows how q_t, the probability assigned to the mean-reverting regime, evolves after a randomly generated earnings path. The parameter values are $\pi_L = 1/3$, $\pi_H = 3/4$, $\lambda_1 = 0.1$, and $\lambda_2 = 0.3$.

t	y_t	q_t	t	y_t	q_t
0	$+y$	0.50	11	$+y$	0.74
1	$-y$	0.80	12	$+y$	0.56
2	$+y$	0.90	13	$+y$	0.44
3	$-y$	0.93	14	$+y$	0.36
4	$+y$	0.94	15	$-y$	0.74
5	$+y$	0.74	16	$+y$	0.89
6	$-y$	0.89	17	$+y$	0.69
7	$-y$	0.69	18	$-y$	0.87
8	$+y$	0.87	19	$+y$	0.92
9	$-y$	0.92	20	$+y$	0.72
10	$+y$	0.94			

Economic message. Alternating signs raise q_t because reversals look like evidence for Model 1. Same-sign streaks lower q_t because they look like evidence for Model 2. Thus the same signal can be interpreted differently depending on the recent sequence.

5.2 Figure 1: parameter regions for both effects

Figure note. The original paper plots the parameter regions visually. The study note summarizes the figure rather than reproducing the image.

Read. The top panel marks (π_L, π_H) pairs that satisfy sufficient conditions for both underreaction and overreaction when $\lambda_1 = 0.1$ and $\lambda_2 = 0.3$. The bottom panels separately show regions that generate only overreaction or only underreaction.

Economic message. The joint result is not automatic. If both π_L and π_H are high, the investor expects continuation too often, so overreaction dominates. If both are low, the investor expects reversal too often, so underreaction dominates. The model needs a middle configuration: one regime mean-reverts, the other trends.

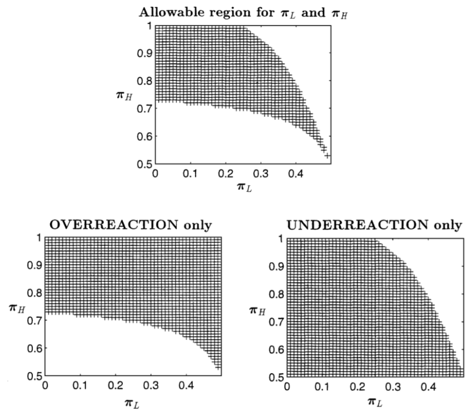

Fig. 1. Shaded area in graph at top marks the $[\pi_L, \pi_H]$ pairs which satisfy the sufficient conditions for both underreaction and overreaction, when $\lambda_1 = 0.1$ and $\lambda_2 = 0.3$. Graph at bottom left (right) shows the $[\pi_L, \pi_H]$ pairs that satisfy the condition for overreaction (underreaction) only. π_L (π_H) is the probability, in the mean-reverting (trending) regime, that next period's earnings shock will be of the same sign as last period's earnings shock. λ_1 and λ_2 govern the transition probabilities between regimes.

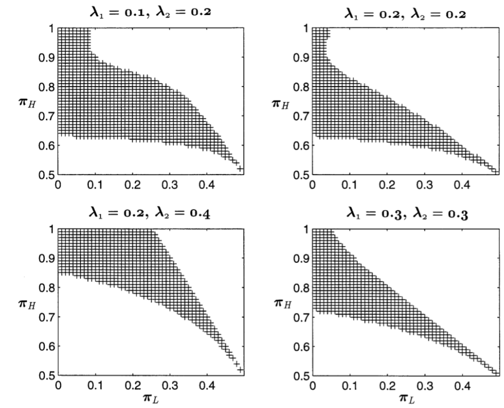

Fig. 2. Shaded area shows the $[\pi_L, \pi_H]$ pairs which satisfy the sufficient conditions for both underreaction and overreaction for a variety of different values of λ_1 and λ_2. π_L (π_H) is the probability, in the mean-reverting (trending) regime, that next period's earnings shock will be of the same sign as last period's earnings shock. λ_1 and λ_2 govern the transition probabilities between regimes.

5.3 Figure 2: robustness to regime-switching parameters

Figure note. The original paper repeats the parameter-region exercise for alternative regime-switching probabilities. The study note summarizes the figure rather than reproducing the image.

Read. Figure 2 repeats the sufficient-condition exercise for several combinations of λ_1 and λ_2.

Economic message. The result is not driven by one precise regime-switching pair. Across several values of λ_1 and λ_2, there remain meaningful regions of (π_L, π_H) space in which both anomalies arise.

5.4 Table 2: earnings-sorted simulations

Read. The simulation forms portfolios based on n consecutive years of positive or negative earnings changes and reports the next-year return difference between the positive-history and negative-history portfolios.

Earnings-history sort	Next-year return difference
$r^1_+ - r^1_-$	0.0391
$r^2_+ - r^2_-$	0.0131
$r^3_+ - r^3_-$	-0.0072
$r^4_+ - r^4_-$	-0.0309

Economic message. One positive earnings change predicts higher future returns, which is underreaction. As the positive sequence lengthens, the return difference falls and turns negative, which is overreaction.

5.5 Table 3: return-sorted simulations

Read. The simulation sorts firms into deciles based on cumulative returns over the past n years and reports next-year returns of past winners minus past losers.

Return-history sort	Next-year winner-minus-loser return
$r_W^1 - r_L^1$	0.0280
$r_W^2 - r_L^2$	0.0102
$r_W^3 - r_L^3$	-0.0094
$r_W^4 - r_L^4$	-0.0181

Economic message. The model reproduces momentum at short horizons and reversal at longer horizons. This is exactly the joint pattern the paper wants to explain.

5.6 Earnings-price sort

Read. The paper also sorts stocks by earnings-price ratios. The high-E/P portfolio outperforms the low-E/P portfolio by

$$r_{E/P}^{hi} - r_{E/P}^{lo} = 0.0435. \tag{24}$$

Economic message. The model generates a value premium without risk aversion or risk compensation. High-E/P stocks are stocks whose prices are depressed relative to earnings because investors have overreacted pessimistically; low-E/P stocks are priced too optimistically.

5.7 Event-study discussion

Read. The paper discusses whether the model can account for event-study drift beyond earnings announcements. It can naturally explain underreaction to isolated events such as dividend cuts, repurchases, or other corporate announcements if investors treat each event as low-strength information and update too little.

Important nuance. The model is less successful for cases where a firm has already had a long growth history but still shows underreaction to new quarterly earnings surprises. The authors suggest that a richer model could let investors estimate earnings levels and earnings growth separately.

Economic message. The model is powerful but not final. Its main achievement is to show how one belief mechanism can jointly produce underreaction and overreaction, not to mechanically fit every anomaly.

6 Short summary

Barberis, Shleifer, and Vishny build a model in which true earnings follow a random walk but investors mistakenly think earnings switch between a mean-reverting regime and a trending regime. Investors use Bayes' rule within this wrong model. After isolated news, they often put too much probability on mean reversion and underreact. After a long sequence of similar news, they put too much probability on trend continuation and overreact. This produces both short-run return continuation and long-run reversal. Simulations reproduce the qualitative signs of post-earnings drift, momentum, reversal, and value effects.

7 Conclusion

7.1 Core conclusion

A single model of investor belief formation can generate both underreaction and overreaction. The key is that investors are conservative after isolated signals but become representativeness-driven after repeated similar signals.

7.2 Economic intuition

The investor asks the wrong question. Instead of asking, “What is the best forecast if earnings are a random walk?”, he asks, “Am I in a mean-reverting regime or a trending regime?” That mistaken framing causes him to underweight some signals and overweight patterns.

7.3 Takeaway

The paper’s central contribution is not a new anomaly. It is a disciplined behavioral mechanism that connects several anomalies. Short-run momentum and long-run reversal can coexist because investors underreact to individual news but overreact to streaks.